

DDA3020 Homework 3

Due date: May 5, 2026

Instructions

- The **deadline** is **23:59, May 5, 2026**, please submit in time.
- The total score for this assignment is **100 points**.
- **Electronic submission:** Submit your solutions via Blackboard with the following files:
 - One PDF file containing all written answers and programming results:
`DDA3020HW3_studentID_name.pdf`
 - Two Python scripts for the programming question:
`HW3_name_Q2_1.ipynb`, `HW3_name_Q2_2.py`

Replace `studentID` with your student ID and `name` with your name.

- The complete and executable code must be submitted. If you only report results in the PDF but do not submit the source code (`.py` or `.ipynb` file), you will receive **0 points** for the programming question.
- Answer the questions in English. Otherwise, you will lose half of the points.
- Collaboration policy: You need to solve all questions independently and collaboration between students is **NOT** allowed.

1 Written Problems (50 points)

1.1 An EM algorithm for a Mixture of Bernoullis, (25 points)

In this section, you will derive an expectation-maximization (EM) algorithm to cluster black and white images. The inputs $\mathbf{x}^{(i)}$ can be thought of as vectors of binary values corresponding to black and white pixel values, and the goal is to cluster the images into groups. You will be using a mixture of Bernoullis model to tackle this problem.

For the sake of brevity, you do not need to substitute in previously derived expression in later problems. For example, beyond question 1.1.1 you may use $P(\mathbf{x}^{(i)} | \mathbf{p}^{(k)})$ in your answers.

1.1.1 Mixture of Bernoullis

- (2pts) Consider a vector of binary random variables, $\mathbf{x} \in \{0, 1\}^D$. Assume each variable x_d is drawn from a Bernoulli (p_d) distribution, so $P(x_d = 1) = p_d$. Let $\mathbf{p} \in (0, 1)^D$ be the resulting vector of Bernoulli parameters. Write an expression for $P(\mathbf{x} | \mathbf{p})$.
- (2pts) Now suppose we have a mixture of K Bernoulli distributions: each vector $\mathbf{x}^{(i)}$ is drawn from some vector of Bernoulli random variables with parameters $\mathbf{p}^{(k)}$, we will call this Bernoulli ($\mathbf{p}^{(k)}$). Let $\mathbf{p} = \{\mathbf{p}^{(1)}, \dots, \mathbf{p}^{(K)}\}$. $\boldsymbol{\pi} = \{\pi_1, \dots, \pi_K\}$. Assume a distribution π_k over the selection of which set of Bernoulli parameters $\mathbf{p}^{(k)}$ is chosen. Write an expression for $P(\mathbf{x}^{(i)} | \mathbf{p}, \boldsymbol{\pi})$.
- (2pts) Finally, suppose we have i.i.d. inputs $\mathbf{X} = \{\mathbf{x}^{(i)}\}_{i=1 \dots N}$. Using the above, write an expression for the log likelihood of the data $\mathbf{X}$, $\log P(\mathbf{X} | \boldsymbol{\pi}, \mathbf{p})$.

1.1.2 Expectation step

- (4pts) Now, we introduce the latent variables for the EM algorithm. Let $\mathbf{z}^{(i)} \in \{0, 1\}^K$ be an indicator vector, such that $z_k^{(i)} = 1$ if $\mathbf{x}^{(i)}$ was drawn from a Bernoulli ($\mathbf{p}^{(k)}$), and 0 otherwise. Let $\mathbf{Z} = \{\mathbf{z}^{(i)}\}_{i=1 \dots N}$. What is $P(\mathbf{z}^{(i)} | \boldsymbol{\pi})$? What is $P(\mathbf{x}^{(i)} | \mathbf{z}^{(i)}, \mathbf{p}, \boldsymbol{\pi})$?
- (2pts) Using the above two quantities, derive the likelihood of the data and the latent variables, $P(\mathbf{Z}, \mathbf{X} | \boldsymbol{\pi}, \mathbf{p})$.
- (5pts) Let $\eta(z_k^{(i)}) = E[z_k^{(i)} | \mathbf{x}^{(i)}, \boldsymbol{\pi}, \mathbf{p}]$. Show that

$$\eta(z_k^{(i)}) = \frac{\pi_k \prod_{d=1}^D (p_d^{(k)})^{x_d^{(i)}} (1 - p_d^{(k)})^{1-x_d^{(i)}}}{\sum_j \pi_j \prod_{d=1}^D (p_d^{(j)})^{x_d^{(i)}} (1 - p_d^{(j)})^{1-x_d^{(i)}}}$$

Let $\tilde{\mathbf{p}}, \tilde{\boldsymbol{\pi}}$ be the new parameters that we'd like to maximize, so $\mathbf{p}, \boldsymbol{\pi}$ are from the previous

iteration. Use this to derive the following final expression for the E step in the expectation-maximization algorithm:

$$E[\log P(\mathbf{Z}, \mathbf{X} \mid \tilde{\mathbf{p}}, \tilde{\boldsymbol{\pi}}) \mid \mathbf{X}, \mathbf{p}, \boldsymbol{\pi}] = \sum_{i=1}^N \sum_{k=1}^K \eta(z_k^{(i)}) \left[\log \tilde{\pi}_k + \sum_{d=1}^D \left(x_d^{(i)} \log \tilde{p}_d^{(k)} + (1 - x_d^{(i)}) \log (1 - \tilde{p}_d^{(k)}) \right) \right]$$

1.1.3 Maximization step

1. (4pts) We need to maximize the above expression with respect to $\tilde{\boldsymbol{\pi}}, \tilde{\mathbf{p}}$. First, show that the value of $\tilde{\mathbf{p}}$ that maximizes the E step is

$$\tilde{\mathbf{p}}^{(k)} = \frac{\sum_{i=1}^N \eta(z_k^{(i)}) \mathbf{x}^{(i)}}{N_k}$$

where $N_k = \sum_{i=1}^N \eta(z_k^{(i)})$.

2. (4pts) Show that the value of $\tilde{\boldsymbol{\pi}}$ that maximizes the E step is

$$\tilde{\pi}_k = \frac{N_k}{\sum_{k'} N_{k'}}$$

The exponential families notation may be useful. Alternatively, you can use Lagrange multipliers.

Solution:

In this section, you will derive an expectation-maximization (EM) algorithm to cluster black and white images. The inputs $\mathbf{x}^{(i)}$ can be thought of as vectors of binary values corresponding to black and white pixel values, and the goal is to cluster the images into groups. You will be using a mixture of Bernoullis model to tackle this problem.

For the sake of brevity, you do not need to substitute in previously derived expression in later problems. For example, beyond question 1.1.1 you may use $P(\mathbf{x}^{(i)} \mid \mathbf{p}^{(k)})$ in your answers.

1.1.4 Mixture of Bernoullis

1. (2pts) Consider a vector of binary random variables, $\mathbf{x} \in \{0, 1\}^D$. Assume each variable x_d is drawn from a Bernoulli (p_d) distribution, so $P(x_d = 1) = p_d$. Let $\mathbf{p} \in (0, 1)^D$ be the resulting vector of Bernoulli parameters. Write an expression for $P(\mathbf{x} \mid \mathbf{p})$.

Solution:

$$P(\mathbf{x} | \mathbf{p}) = \prod_{d=1}^D p_d^{x_d} (1 - p_d)^{1-x_d}$$

2. (2pts) Now suppose we have a mixture of K Bernoulli distributions: each vector $\mathbf{x}^{(i)}$ is drawn from some vector of Bernoulli random variables with parameters $\mathbf{p}^{(k)}$, we will call this Bernoulli ($\mathbf{p}^{(k)}$). Let $\mathbf{p} = \{\mathbf{p}^{(1)}, \dots, \mathbf{p}^{(K)}\}$. $\boldsymbol{\pi} = \{\pi_1, \dots, \pi_K\}$. Assume a distribution π_k over the selection of which set of Bernoulli parameters $\mathbf{p}^{(k)}$ is chosen. Write an expression for $P(\mathbf{x}^{(i)} | \mathbf{p}, \boldsymbol{\pi})$.

Solution: Let A_k be the event that $\mathbf{x} = \mathbf{x}^{(i)}$ was drawn from $\mathbf{p}^{(k)}$. Then:

$$P(\mathbf{x} | \mathbf{p}, \boldsymbol{\pi}) = \sum_k P(\mathbf{x}, A_k | \mathbf{p}, \boldsymbol{\pi}) = \sum_k P(\mathbf{x} | A_k, \mathbf{p}, \boldsymbol{\pi}) P(A_k | \mathbf{p}, \boldsymbol{\pi}) = \sum_k \pi_k P(\mathbf{x} | \mathbf{p}^{(k)})$$

3. (2pts) Finally, suppose we have i.i.d. inputs $\mathbf{X} = \{\mathbf{x}^{(i)}\}_{i=1 \dots N}$. Using the above, write an expression for the log likelihood of the data $\mathbf{X}$, $\log P(\mathbf{X} | \boldsymbol{\pi}, \mathbf{p})$.

Solution:

$$\log L(\mathbf{p}, \boldsymbol{\pi}) = \sum_{i=1}^N \log P(\mathbf{x}^{(i)} | \mathbf{p}, \boldsymbol{\pi})$$

1.1.5 Expectation step

1. (4pts) Now, we introduce the latent variables for the EM algorithm. Let $\mathbf{z}^{(i)} \in \{0, 1\}^K$ be an indicator vector, such that $z_k^{(i)} = 1$ if $\mathbf{x}^{(i)}$ was drawn from a Bernoulli ($\mathbf{p}^{(k)}$), and 0 otherwise. Let $\mathbf{Z} = \{\mathbf{z}^{(i)}\}_{i=1 \dots N}$. What is $P(\mathbf{z}^{(i)} | \boldsymbol{\pi})$? What is $P(\mathbf{x}^{(i)} | \mathbf{z}^{(i)}, \mathbf{p}, \boldsymbol{\pi})$?

Solution: Let $A_k^{(i)}$ be the event that $\mathbf{x}^{(i)}$ was drawn from $\mathbf{p}^{(k)}$.

$$P(\mathbf{z}^{(i)} | \boldsymbol{\pi}) = \prod_{k=1}^K \pi_k^{z_k^{(i)}} \text{ 2 points}$$

$$P(\mathbf{x}^{(i)} | \mathbf{z}^{(i)}, \mathbf{p}, \boldsymbol{\pi}) = \prod_{k=1}^K P(\mathbf{x}^{(i)} | \mathbf{z}^{(i)}, \mathbf{p}, \boldsymbol{\pi}, A_k^{(i)})^{z_k^{(i)}} = \prod_{k=1}^K \left[P(\mathbf{x}^{(i)} | \mathbf{p}^{(k)}) \right]^{z_k^{(i)}} \text{ 2 points}$$

2. (2pts) Using the above two quantities, derive the likelihood of the data and the latent variables, $P(\mathbf{Z}, \mathbf{X} | \boldsymbol{\pi}, \mathbf{p})$.

Solution:

$$P(\mathbf{Z}, \mathbf{X} \mid \boldsymbol{\pi}, \mathbf{p}) = \prod_{i=1}^N P\left(\mathbf{x}^{(i)}, \mathbf{z}^{(i)} \mid \boldsymbol{\pi}, \mathbf{p}\right) = \prod_{i=1}^N P\left(\mathbf{x}^{(i)} \mid \mathbf{z}^{(i)}, \boldsymbol{\pi}, \mathbf{p}\right) P\left(\mathbf{z}^{(i)} \mid \boldsymbol{\pi}\right) = \prod_{i=1}^N \left[\prod_{k=1}^K \left[P\left(\mathbf{x}^{(i)} \mid \mathbf{p}^{(k)}\right) \right]^{z_k^{(i)}} \right]$$

3. (5pts) Let $\eta\left(z_k^{(i)}\right) = E\left[z_k^{(i)} \mid \mathbf{x}^{(i)}, \boldsymbol{\pi}, \mathbf{p}\right]$. Show that

$$\eta\left(z_k^{(i)}\right) = \frac{\pi_k \prod_{d=1}^D \left(p_d^{(k)}\right)^{x_d^{(i)}} \left(1 - p_d^{(k)}\right)^{1-x_d^{(i)}}}{\sum_j \pi_j \prod_{d=1}^D \left(p_d^{(j)}\right)^{x_d^{(i)}} \left(1 - p_d^{(j)}\right)^{1-x_d^{(i)}}}$$

Let $\tilde{\mathbf{p}}, \tilde{\boldsymbol{\pi}}$ be the new parameters that we'd like to maximize, so $\mathbf{p}, \boldsymbol{\pi}$ are from the previous iteration. Use this to derive the following final expression for the E step in the expectation-maximization algorithm:

$$E[\log P(\mathbf{Z}, \mathbf{X} \mid \tilde{\mathbf{p}}, \tilde{\boldsymbol{\pi}}) \mid \mathbf{X}, \mathbf{p}, \boldsymbol{\pi}] = \sum_{i=1}^N \sum_{k=1}^K \eta\left(z_k^{(i)}\right) \left[\log \tilde{\pi}_k + \sum_{d=1}^D \left(x_d^{(i)} \log \tilde{p}_d^{(k)} + \left(1 - x_d^{(i)}\right) \log \left(1 - \tilde{p}_d^{(k)}\right) \right) \right]$$

Solution:

$$\begin{aligned} \eta\left(z_k^{(i)}\right) &= E\left[z_k^{(i)} \mid \mathbf{x}^{(i)}, \boldsymbol{\pi}, \mathbf{p}\right] \\ &= P\left[z_k^{(i)} = 1 \mid \mathbf{x}^{(i)}, \boldsymbol{\pi}, \mathbf{p}\right] \\ &= \frac{P\left(\mathbf{x}^{(i)} \mid z_k^{(i)} = 1, \boldsymbol{\pi}, \mathbf{p}\right) P\left(z_k^{(i)} = 1 \mid \boldsymbol{\pi}, \mathbf{p}\right)}{\sum_{k'} P\left(\mathbf{x}^{(i)} \mid z_{k'}^{(i)} = 1, \boldsymbol{\pi}, \mathbf{p}\right) P\left(z_{k'}^{(i)} = 1 \mid \boldsymbol{\pi}, \mathbf{p}\right)} \\ &= \frac{\pi_k \prod_{d=1}^D \left(p_d^{(k)}\right)^{x_d^{(i)}} \left(1 - p_d^{(k)}\right)^{1-x_d^{(i)}}}{\sum_{k'} \pi_{k'} \prod_{d=1}^D \left(p_d^{(k')}\right)^{x_d^{(i)}} \left(1 - p_d^{(k')}\right)^{1-x_d^{(i)}}} \quad \text{2 points} \end{aligned}$$

Next we compute the log likelihood:

$$\begin{aligned}
\log P(\mathbf{Z}, \mathbf{X} \mid \tilde{\boldsymbol{\pi}}, \tilde{\mathbf{p}}) &= \sum_{i=1}^N \left[\sum_{k=1}^K z_k^{(i)} \log \left[P(\mathbf{x}^{(i)} \mid \tilde{\mathbf{p}}^{(k)}) \right] \right] + \left[\sum_{k=1}^K z_k^{(i)} \log \tilde{\pi}_k \right] \\
&= \sum_{i=1}^N \sum_{k=1}^K z_k^{(i)} \left[\log P(\mathbf{x}^{(i)} \mid \tilde{\mathbf{p}}^{(k)}) + \log \tilde{\pi}_k \right] \\
&= \sum_{i=1}^N \sum_{k=1}^K z_k^{(i)} \left[\log \tilde{\pi}_k + \log \prod_{d=1}^D \left(\tilde{p}_d^{(k)} \right)^{x_d^{(i)}} \left(1 - \tilde{p}_d^{(k)} \right)^{1-x_d^{(i)}} \right] \\
&= \sum_{i=1}^N \sum_{k=1}^K z_k^{(i)} \left[\log \tilde{\pi}_k + \sum_{d=1}^D \left(x_d^{(i)} \log \left(\tilde{p}_d^{(k)} \right) + \left(1 - x_d^{(i)} \right) \log \left(1 - \tilde{p}_d^{(k)} \right) \right) \right]
\end{aligned}$$

Taking the expected value and replacing $E \left[z_k^{(i)} \right] = \eta \left(z_k^{(i)} \right)$ finishes the solution. **3 points**

1.1.6 Maximization step

- (4pts) We need to maximize the above expression with respect to $\tilde{\boldsymbol{\pi}}, \tilde{\mathbf{p}}$. First, show that the value of $\tilde{\mathbf{p}}$ that maximizes the E step is

$$\tilde{\mathbf{p}}^{(k)} = \frac{\sum_{i=1}^N \eta \left(z_k^{(i)} \right) \mathbf{x}^{(i)}}{N_k}$$

where $N_k = \sum_{i=1}^N \eta \left(z_k^{(i)} \right)$.

Solution: Setting the derivative to 0 :

$$\frac{d}{d\tilde{p}_d^{(k)}} E[\log P(\mathbf{Z}, \mathbf{X} \mid \tilde{\boldsymbol{\pi}}, \tilde{\mathbf{p}})] = \sum_{i=1}^N \eta \left(z_k^{(i)} \right) \left[\frac{x_d^{(i)}}{\tilde{p}_d^{(k)}} - \frac{1 - x_d^{(i)}}{1 - \tilde{p}_d^{(k)}} \right] = 0 \quad \text{2 points}$$

Multiply by the denominators:

$$\sum_{i=1}^N \eta \left(z_k^{(i)} \right) \left[x_d^{(i)} \left(1 - \tilde{p}_d^{(k)} \right) - \left(1 - x_d^{(i)} \right) \tilde{p}_d^{(k)} \right] = \sum_{i=1}^N \eta \left(z_k^{(i)} \right) \left[-\tilde{p}_d^{(k)} + x_d^{(i)} \right] = 0$$

Solving for $\tilde{p}_d^{(k)}$ results in

$$\tilde{p}_d^{(k)} = \frac{\sum_{i=1}^N \eta \left(z_k^{(i)} \right) x_d^{(i)}}{\sum_{i=1}^N \eta \left(z_k^{(i)} \right)} = \frac{\sum_{i=1}^N \eta \left(z_k^{(i)} \right) x_d^{(i)}}{N_k} \quad \text{2 points}$$

2. (4pts) Show that the value of $\tilde{\pi}$ that maximizes the E step is

$$\tilde{\pi}_k = \frac{N_k}{\sum_{k'} N_{k'}}$$

The exponential families notation may be useful. Alternatively, you can use Lagrange multipliers.

Solution: We only need to maximize $\sum_{i=1}^N \sum_{k=1}^K \eta(z_k^{(i)}) \log \tilde{\pi}_k$ since the rest is not a function of $\tilde{\pi}$. In order to keep $\tilde{\pi}$ a distribution, we require $\sum_k \tilde{\pi}_k = 1$. Let λ be the dual variable for this constraint:

$$L(\tilde{\pi}, \lambda) = - \sum_{i=1}^N \sum_{k=1}^K \eta(z_k^{(i)}) \log \tilde{\pi}_k + \lambda \left(\sum_{k=1}^K \tilde{\pi}_k - 1 \right)$$

Taking the derivate w.r.t. $\tilde{\pi}_k$:

$$\frac{d}{d\tilde{\pi}_k} L(\tilde{\pi}, \lambda) = - \sum_{i=1}^N \frac{\eta(z_k^{(i)})}{\tilde{\pi}_k} + \lambda = 0$$

solve for $\tilde{\pi}_k$ and we get

$$\tilde{\pi}_k = \frac{\sum_{i=1}^N \eta(z_k^{(i)})}{\lambda} = \frac{N_k}{\lambda} \quad \text{2 points}$$

Now we solve for λ :

$$L(\lambda) = - \sum_{i=1}^N \sum_{k=1}^K \eta(z_k^{(i)}) (\log N_k - \log \lambda) + \left(\sum_{k=1}^K N_k - \lambda \right)$$

Take the derivative w.r.t. λ :

$$\frac{1}{\lambda} \sum_{i=1}^N \sum_{k=1}^K \eta(z_k^{(i)}) - 1 = 0$$

Solve for λ

$$\lambda = \sum_{i=1}^N \sum_{k=1}^K \eta(z_k^{(i)}) = \sum_{k=1}^K N_k \quad \text{2 points}$$

1.2 Manual Calculation, (25 points)

1.2.1 K-Means Calculation, (12 points)

Consider the following 2D dataset with 6 points: $x_1 = (1, 1)$, $x_2 = (1, 2)$, $x_3 = (2, 1)$, $x_4 = (6, 6)$, $x_5 = (6, 7)$, $x_6 = (7, 6)$. Apply the K-means algorithm manually with $K = 2$ clusters and initial centroids at $c_1 = (0, 0)$ and $c_2 = (5, 5)$. Show all steps of the algorithm until convergence, including:

1. (2pts) The assignment of each point to the nearest centroid after the first iteration.
2. (2pts) The calculation of new centroids after the first iteration.
3. (3pts) The assignment of points after the second iteration.
4. (2pts) The final centroids and cluster assignments after convergence.
5. (3pts) Calculate the final value of the objective function $J = \sum_{i=1}^n \sum_{k=1}^K r_{ik} \|x_i - c_k\|^2$.

1.2.2 PCA Calculation, (13 points)

Consider a dataset containing 6 two-dimensional data points: $x_1 = (1, 2)$, $x_2 = (2, 3)$, $x_3 = (3, 2)$, $x_4 = (3, 4)$, $x_5 = (4, 3)$, $x_6 = (5, 4)$. Perform PCA on the given dataset to reduce it to 1 dimension ($K=1$), show all intermediate calculations and final results:

1. (2pts) Calculate the mean vector μ
2. (2pts) Compute the empirical covariance matrix Σ
3. (4pts) Find the eigenvalues and eigenvectors of Σ
4. (2pts) Select the top eigenvector and compute the new 1D representations $z(n)$ for all data points
5. (3pts) Calculate the reconstruction error (sum of squared distances between original points and their reconstructions)

Solution:

K-Means Calculation (12 points)

Initial centroids: $c_1 = (0, 0)$, $c_2 = (5, 5)$

(2 points) First iteration - Assignment step: Calculate Euclidean distances from each point to both centroids:

For $(1, 1)$: $\|(1, 1) - (0, 0)\| = \sqrt{1^2 + 1^2} = \sqrt{2} \approx 1.414$ $\|(1, 1) - (5, 5)\| = \sqrt{4^2 + 4^2} = \sqrt{32} \approx 5.657 \Rightarrow$
Assign to cluster 1

For $(1, 2)$: $\|(1, 2) - (0, 0)\| = \sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.236$ $\|(1, 2) - (5, 5)\| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5.000 \Rightarrow$
Assign to cluster 1

For (2, 1): $\|(2, 1) - (0, 0)\| = \sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.236$ $\|(2, 1) - (5, 5)\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.000 \Rightarrow$
Assign to cluster 1

For (6, 6): $\|(6, 6) - (0, 0)\| = \sqrt{6^2 + 6^2} = \sqrt{72} \approx 8.485$ $\|(6, 6) - (5, 5)\| = \sqrt{1^2 + 1^2} = \sqrt{2} \approx 1.414 \Rightarrow$
Assign to cluster 2

For (6, 7): $\|(6, 7) - (0, 0)\| = \sqrt{6^2 + 7^2} = \sqrt{85} \approx 9.220$ $\|(6, 7) - (5, 5)\| = \sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.236 \Rightarrow$
Assign to cluster 2

For (7, 6): $\|(7, 6) - (0, 0)\| = \sqrt{7^2 + 6^2} = \sqrt{85} \approx 9.220$ $\|(7, 6) - (5, 5)\| = \sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.236 \Rightarrow$
Assign to cluster 2

(2 points) First iteration - Update step:

Cluster 1 contains: (1, 1), (1, 2), (2, 1) $c_1 = \left(\frac{1+1+2}{3}, \frac{1+2+1}{3}\right) = \left(\frac{4}{3}, \frac{4}{3}\right) \approx (1.333, 1.333)$

Cluster 2 contains: (6, 6), (6, 7), (7, 6) $c_2 = \left(\frac{6+6+7}{3}, \frac{6+7+6}{3}\right) = \left(\frac{19}{3}, \frac{19}{3}\right) \approx (6.333, 6.333)$

(2 points) Second iteration - Assignment step:

For (1, 1): $\|(1, 1) - (1.333, 1.333)\| = \sqrt{(0.333)^2 + (0.333)^2} \approx 0.471$ $\|(1, 1) - (6.333, 6.333)\| = \sqrt{(5.333)^2 + (5.333)^2} \approx 7.542 \Rightarrow$ Assign to cluster 1 (unchanged)

For (1, 2): $\|(1, 2) - (1.333, 1.333)\| = \sqrt{(0.333)^2 + (0.667)^2} \approx 0.745$ $\|(1, 2) - (6.333, 6.333)\| = \sqrt{(5.333)^2 + (4.333)^2} \approx 6.867 \Rightarrow$ Assign to cluster 1 (unchanged)

For (2, 1): $\|(2, 1) - (1.333, 1.333)\| = \sqrt{(0.667)^2 + (0.333)^2} \approx 0.745$ $\|(2, 1) - (6.333, 6.333)\| = \sqrt{(4.333)^2 + (5.333)^2} \approx 6.867 \Rightarrow$ Assign to cluster 1 (unchanged)

For (6, 6): $\|(6, 6) - (1.333, 1.333)\| = \sqrt{(4.667)^2 + (4.667)^2} \approx 6.600$ $\|(6, 6) - (6.333, 6.333)\| = \sqrt{(0.333)^2 + (0.333)^2} \approx 0.471 \Rightarrow$ Assign to cluster 2 (unchanged)

For (6, 7): $\|(6, 7) - (1.333, 1.333)\| = \sqrt{(4.667)^2 + (5.667)^2} \approx 7.333$ $\|(6, 7) - (6.333, 6.333)\| = \sqrt{(0.333)^2 + (0.667)^2} \approx 0.745 \Rightarrow$ Assign to cluster 2 (unchanged)

For (7, 6): $\|(7, 6) - (1.333, 1.333)\| = \sqrt{(5.667)^2 + (4.667)^2} \approx 7.333$ $\|(7, 6) - (6.333, 6.333)\| = \sqrt{(0.667)^2 + (0.333)^2} \approx 0.745 \Rightarrow$ Assign to cluster 2 (unchanged)

(1 points) Since all assignments remain unchanged, the algorithm has converged.

(2 points) Final result:

- Final centroids: $c_1 = (1.333, 1.333)$, $c_2 = (6.333, 6.333)$
- Cluster assignments: Points (1, 1), (1, 2), (2, 1) in cluster 1; Points (6, 6), (6, 7), (7, 6) in cluster 2

(1 points) Final objective function value:

For cluster 1: $\|(1, 1) - (1.333, 1.333)\|^2 = (0.333)^2 + (0.333)^2 = 0.222$ $\|(1, 2) - (1.333, 1.333)\|^2 = (0.333)^2 + (0.667)^2 = 0.556$ $\|(2, 1) - (1.333, 1.333)\|^2 = (0.667)^2 + (0.333)^2 = 0.556$ Sum for cluster 1: $0.222 + 0.556 + 0.556 = 1.334$

For cluster 2: $\|(6, 6) - (6.333, 6.333)\|^2 = (0.333)^2 + (0.333)^2 = 0.222$ $\|(6, 7) - (6.333, 6.333)\|^2 = (0.333)^2 + (0.667)^2 = 0.556$ $\|(7, 6) - (6.333, 6.333)\|^2 = (0.667)^2 + (0.333)^2 = 0.556$ Sum for cluster 2: $0.222 + 0.556 + 0.556 = 1.334$

(2 points) Total objective function: $J = 1.334 + 1.334 = 2.668$

PCA Calculation (13 points)

Dataset: $\mathbf{x}^{(1)} = (1, 2)$, $\mathbf{x}^{(2)} = (2, 3)$, $\mathbf{x}^{(3)} = (3, 2)$, $\mathbf{x}^{(4)} = (3, 4)$, $\mathbf{x}^{(5)} = (4, 3)$, $\mathbf{x}^{(6)} = (5, 4)$

(2 points) 1. Calculate mean vector $\boldsymbol{\mu}$:

$$\begin{aligned}\mu_1 &= \frac{1 + 2 + 3 + 3 + 4 + 5}{6} = \frac{18}{6} = 3 \\ \mu_2 &= \frac{2 + 3 + 2 + 4 + 3 + 4}{6} = \frac{18}{6} = 3 \\ \boldsymbol{\mu} &= (3, 3)\end{aligned}$$

(2 points) 2. Compute empirical covariance matrix $\boldsymbol{\Sigma}$:

First, compute centered data: $\mathbf{x}^{(n)} - \boldsymbol{\mu}$

$$\begin{aligned}\mathbf{x}^{(1)} - \boldsymbol{\mu} &= (-2, -1) \\ \mathbf{x}^{(2)} - \boldsymbol{\mu} &= (-1, 0) \\ \mathbf{x}^{(3)} - \boldsymbol{\mu} &= (0, -1) \\ \mathbf{x}^{(4)} - \boldsymbol{\mu} &= (0, 1) \\ \mathbf{x}^{(5)} - \boldsymbol{\mu} &= (1, 0) \\ \mathbf{x}^{(6)} - \boldsymbol{\mu} &= (2, 1)\end{aligned}$$

$$\begin{aligned}\boldsymbol{\Sigma} &= \frac{1}{6} \sum_{n=1}^6 (\mathbf{x}^{(n)} - \boldsymbol{\mu})(\mathbf{x}^{(n)} - \boldsymbol{\mu})^\top \\ &= \frac{1}{6} \left[\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \right] \\ &= \frac{1}{6} \begin{pmatrix} 10 & 4 \\ 4 & 4 \end{pmatrix} \\ &= \begin{pmatrix} \frac{5}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} \end{pmatrix}\end{aligned}$$

3. Find eigenvalues and eigenvectors of Σ :

(2 points) Solve $\det(\Sigma - \lambda \mathbf{I}) = 0$:

$$\begin{aligned}
& \det \begin{pmatrix} \frac{5}{3} - \lambda & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} - \lambda \end{pmatrix} = 0 \\
& \left(\frac{5}{3} - \lambda\right) \left(\frac{2}{3} - \lambda\right) - \left(\frac{2}{3}\right)^2 = 0 \\
& \lambda^2 - \frac{7}{3}\lambda + \frac{2}{3} = 0 \\
& 3\lambda^2 - 7\lambda + 2 = 0 \\
& (3\lambda - 1)(\lambda - 2) = 0 \\
& \lambda_1 = 2, \quad \lambda_2 = \frac{1}{3}
\end{aligned}$$

(1 points) For $\lambda_1 = 2$:

$$\begin{aligned}
& (\Sigma - 2\mathbf{I})\mathbf{v}_1 = \mathbf{0} \\
& \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{4}{3} \end{pmatrix} \mathbf{v}_1 = \mathbf{0} \\
& -\frac{1}{3}v_{11} + \frac{2}{3}v_{12} = 0 \Rightarrow v_{11} = 2v_{12} \\
& \mathbf{v}_1 = (2, 1) \\
& \|\mathbf{v}_1\| = \sqrt{4 + 1} = \sqrt{5} \\
& \mathbf{u}_1 = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right) \approx (0.894, 0.447)
\end{aligned}$$

(1 points) For $\lambda_2 = \frac{1}{3}$:

$$\begin{aligned}
& \left(\Sigma - \frac{1}{3}\mathbf{I}\right)\mathbf{v}_2 = \mathbf{0} \\
& \begin{pmatrix} \frac{4}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{pmatrix} \mathbf{v}_2 = \mathbf{0} \\
& \frac{4}{3}v_{21} + \frac{2}{3}v_{22} = 0 \Rightarrow v_{22} = -2v_{21} \\
& \mathbf{v}_2 = (1, -2) \\
& \|\mathbf{v}_2\| = \sqrt{1 + 4} = \sqrt{5} \\
& \mathbf{u}_2 = \left(\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}\right) \approx (0.447, -0.894)
\end{aligned}$$

(2 points) **4. Select top eigenvector and compute new representations:**

For $K = 1$, we use $\mathbf{u}_1 = (0.894, 0.447)$

$$z^{(n)} = \mathbf{u}_1^\top (\mathbf{x}^{(n)} - \boldsymbol{\mu})$$

$$z^{(1)} = (0.894, 0.447) \cdot (-2, -1) = -1.788 - 0.447 = -2.235$$

$$z^{(2)} = (0.894, 0.447) \cdot (-1, 0) = -0.894 + 0 = -0.894$$

$$z^{(3)} = (0.894, 0.447) \cdot (0, -1) = 0 - 0.447 = -0.447$$

$$z^{(4)} = (0.894, 0.447) \cdot (0, 1) = 0 + 0.447 = 0.447$$

$$z^{(5)} = (0.894, 0.447) \cdot (1, 0) = 0.894 + 0 = 0.894$$

$$z^{(6)} = (0.894, 0.447) \cdot (2, 1) = 1.788 + 0.447 = 2.235$$

5. Calculate reconstruction error:

(1 points) Reconstructions: $\tilde{\mathbf{x}}^{(n)} = \boldsymbol{\mu} + z^{(n)}\mathbf{u}_1$

$$\tilde{\mathbf{x}}^{(1)} = (3, 3) + (-2.235)(0.894, 0.447) = (3, 3) + (-2.00, -1.00) = (1.00, 2.00)$$

$$\tilde{\mathbf{x}}^{(2)} = (3, 3) + (-0.894)(0.894, 0.447) = (3, 3) + (-0.80, -0.40) = (2.20, 2.60)$$

$$\tilde{\mathbf{x}}^{(3)} = (3, 3) + (-0.447)(0.894, 0.447) = (3, 3) + (-0.40, -0.20) = (2.60, 2.80)$$

$$\tilde{\mathbf{x}}^{(4)} = (3, 3) + (0.447)(0.894, 0.447) = (3, 3) + (0.40, 0.20) = (3.40, 3.20)$$

$$\tilde{\mathbf{x}}^{(5)} = (3, 3) + (0.894)(0.894, 0.447) = (3, 3) + (0.80, 0.40) = (3.80, 3.40)$$

$$\tilde{\mathbf{x}}^{(6)} = (3, 3) + (2.235)(0.894, 0.447) = (3, 3) + (2.00, 1.00) = (5.00, 4.00)$$

(1 points) Reconstruction errors:

$$\|\mathbf{x}^{(1)} - \tilde{\mathbf{x}}^{(1)}\|^2 = \|(1, 2) - (1, 2)\|^2 = 0$$

$$\|\mathbf{x}^{(2)} - \tilde{\mathbf{x}}^{(2)}\|^2 = \|(2, 3) - (2.2, 2.6)\|^2 = \|(-0.2, 0.4)\|^2 = 0.04 + 0.16 = 0.20$$

$$\|\mathbf{x}^{(3)} - \tilde{\mathbf{x}}^{(3)}\|^2 = \|(3, 2) - (2.6, 2.8)\|^2 = \|(0.4, -0.8)\|^2 = 0.16 + 0.64 = 0.80$$

$$\|\mathbf{x}^{(4)} - \tilde{\mathbf{x}}^{(4)}\|^2 = \|(3, 4) - (3.4, 3.2)\|^2 = \|(-0.4, 0.8)\|^2 = 0.16 + 0.64 = 0.80$$

$$\|\mathbf{x}^{(5)} - \tilde{\mathbf{x}}^{(5)}\|^2 = \|(4, 3) - (3.8, 3.4)\|^2 = \|(0.2, -0.4)\|^2 = 0.04 + 0.16 = 0.20$$

$$\|\mathbf{x}^{(6)} - \tilde{\mathbf{x}}^{(6)}\|^2 = \|(5, 4) - (5, 4)\|^2 = 0$$

(1 points) Total reconstruction error:

$$J = \frac{1}{6}(0 + 0.20 + 0.80 + 0.80 + 0.20 + 0) = \frac{2.00}{6} = 0.333$$

2 Programming (50 points)

2.1 (PCA: Noise Filtering and Eigenfaces, 25 points)

See HW3_Q2_1.ipynb

You should write the key code blocks you filled in, the output results you obtained, and the required analysis in this PDF file. Note that the results generated by your source code must exactly match those reported in this file. You also should submit the filled `HW3_Q2_1.ipynb` file with outputs visible.

Solution:

See `HW3_Q2_1_sol.ipynb`

2.2 (K-means Implementation and Application, 25 points)**2.2.1 Part 1: Implementing K-means from Scratch**

Implement the K-means clustering algorithm from scratch without using scikit-learn's KMeans class. Your implementation should include the following functions:

1. `initialize_centroids(X, k, random_state=42)`: Randomly initialize k centroids from the data points.
2. `assign_clusters(X, centroids)`: Assign each data point to the nearest centroid.
3. `update_centroids(X, labels, k)`: Update centroids by computing the mean of all points assigned to each cluster.
4. `kmeans_from_scratch(X, k, max_iter=100, tol=1e-4, random_state=42)`: The main K-means function that iteratively assigns clusters and updates centroids until convergence or maximum iterations.

Requirements:

- Use Euclidean distance for computing distances between points and centroids.
- Implement convergence checking based on the change in centroids' positions (use tolerance parameter `tol`).
- Return the final cluster labels, centroids, and the number of iterations performed.
- Set a random seed for reproducibility.

2.2.2 Part 2: Using scikit-learn and Visualization

Use scikit-learn's KMeans implementation to cluster the same dataset. Compare your implementation with scikit-learn's results.

Requirements:

- Generate a synthetic dataset using `sklearn.datasets.make_blobs` with 3 centers, 300 samples, and 2 features.
- Preprocess the data by standardizing it using `StandardScaler`.
- Apply your K-means implementation and scikit-learn's `KMeans` (with the same parameters) to the dataset.
- Create a figure with two subplots side by side:
 - Left subplot: Your implementation's clustering results
 - Right subplot: scikit-learn's clustering results
- Each subplot should show:
 - Data points colored by their cluster assignment
 - Centroids marked with a distinct symbol (e.g., 'X')
 - Appropriate titles, axis labels, and legends
- Include a brief analysis (2-3 sentences) comparing the results from both implementations.

2.2.3 Part 3: Determining Optimal Number of Clusters

Implement methods to determine the optimal number of clusters for a dataset.

Requirements:

- Generate another synthetic dataset using `make_blobs` with 5 centers, 500 samples, and 2 features.
- Standardize the data.
- Implement the elbow method by:
 - Running K-means for k values from 1 to 10
 - Computing the inertia (sum of squared distances to nearest centroid) for each k
 - Plotting inertia vs. number of clusters
- Implement silhouette analysis by:
 - Computing silhouette scores for k values from 2 to 10
 - Plotting silhouette scores vs. number of clusters
 - Creating silhouette plots for k=3, k=5, and k=7

- Based on both methods, determine the optimal number of clusters and justify your choice.
- Apply K-means with the optimal k to the dataset and visualize the results with centroids clearly marked.

2.2.4 Submission Guidelines

- Submit a single Python script (`HW3_Q2_2.py` or `HW3_Q2_2.ipynb`) containing all your code.
- Include comments explaining your code and thought process.
- Save all generated figures as PNG files with appropriate names.
- Include a brief report with your analysis from Parts 2 and 3.
- Ensure your code runs without errors when executed.

2.2.5 Grading Rubric

- **Part 1 (5 points):** Correct implementation of all functions, proper convergence checking, reproducibility
- **Part 2 (8 points):** Correct use of scikit-learn, proper data preprocessing, quality of visualizations, meaningful analysis
- **Part 3 (8 points):** Correct implementation of elbow method and silhouette analysis, proper visualization, justified choice of optimal k
- **Code Quality (4 points):** Readability, comments, organization

Solution: See `HW3_Q2_2_sol.py`